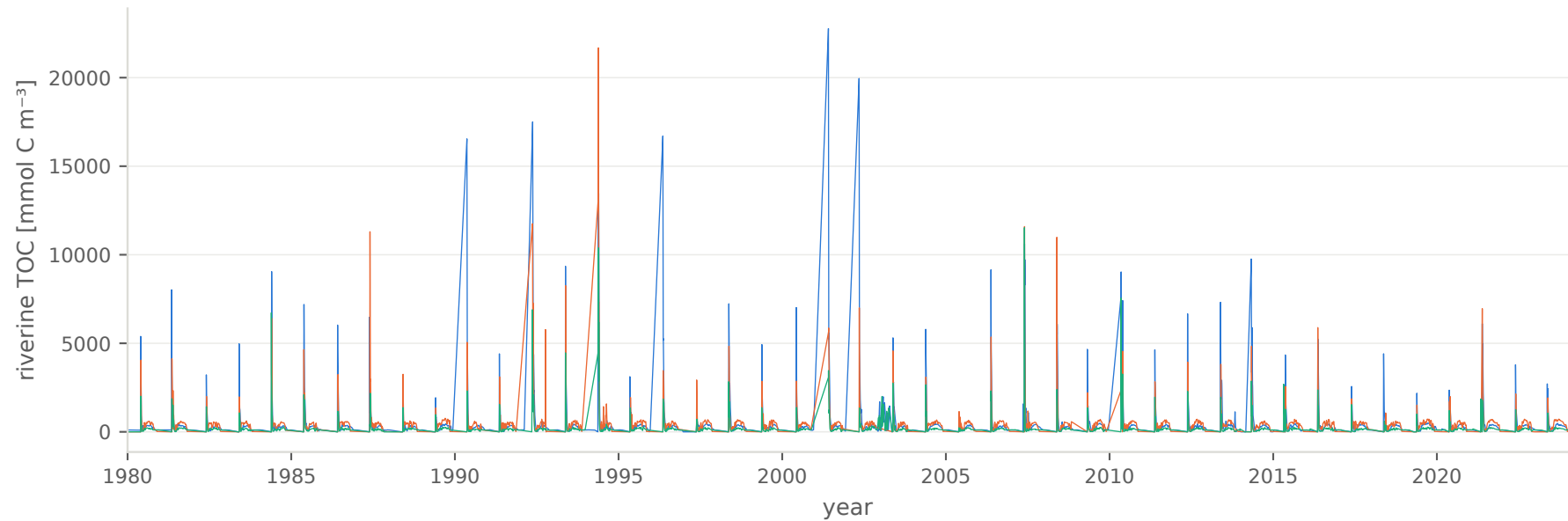
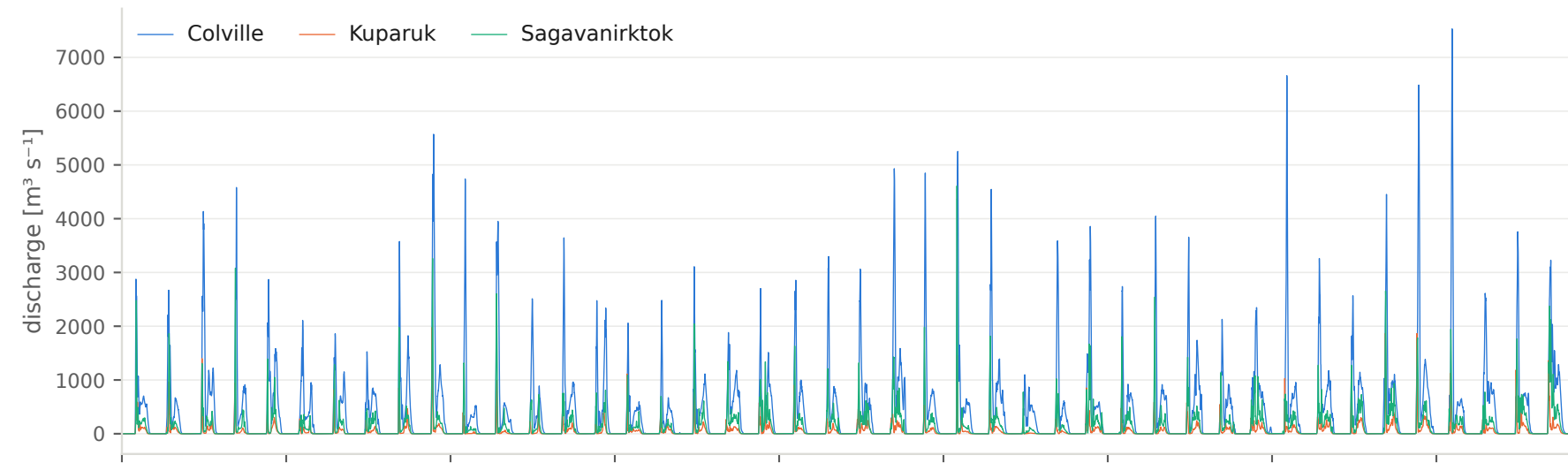


Interannual forcing, 1980-2023 (PWBM-derived)

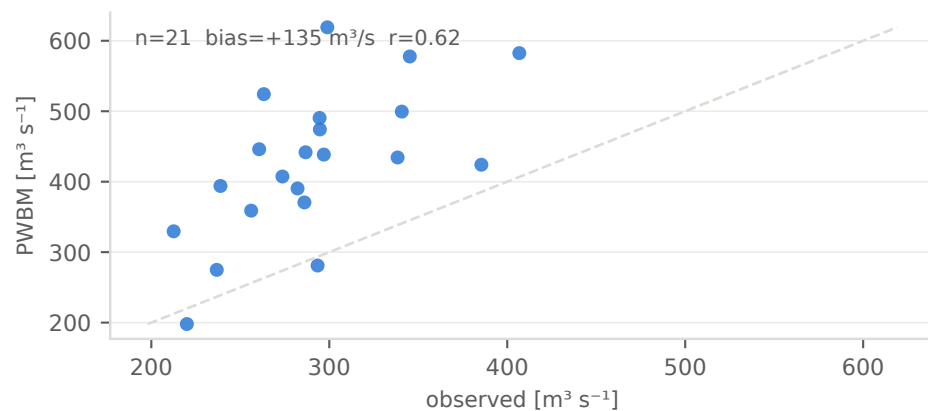
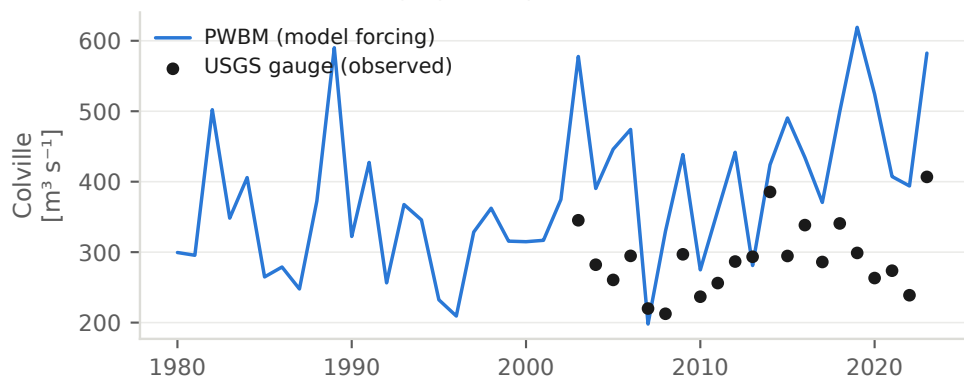
Discharge and riverine TOC (from PWBM DOC — see CLAUDE.md → "Interannual forcing") for the three runnable rivers. Everything else (met, tides, marine b



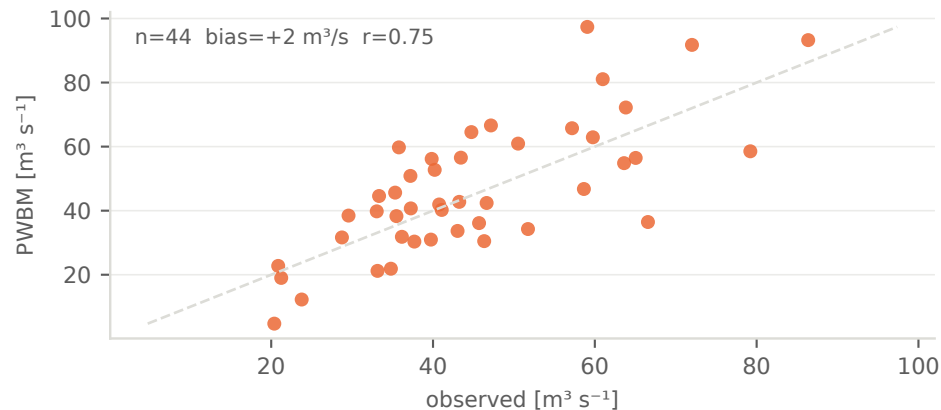
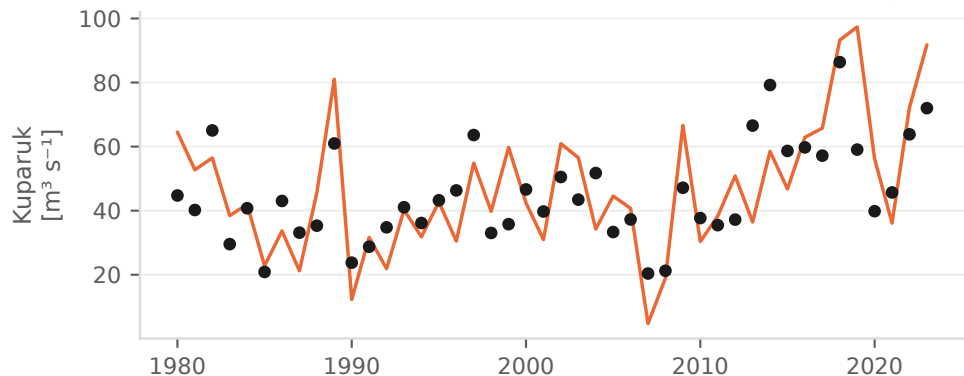
Discharge forcing vs. observation, 1980-2023

PWBM-derived discharge (the model's actual forcing, solid line) against the real USGS gauge annual mean (points), same three gauges tools/fetch_discharge.py uses

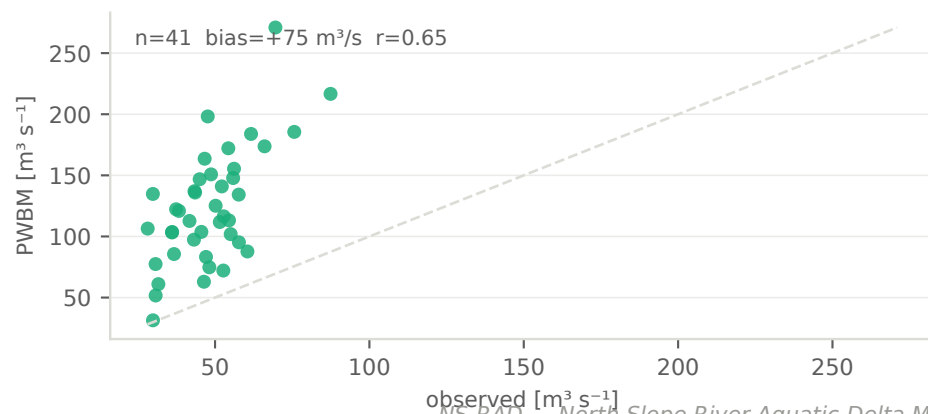
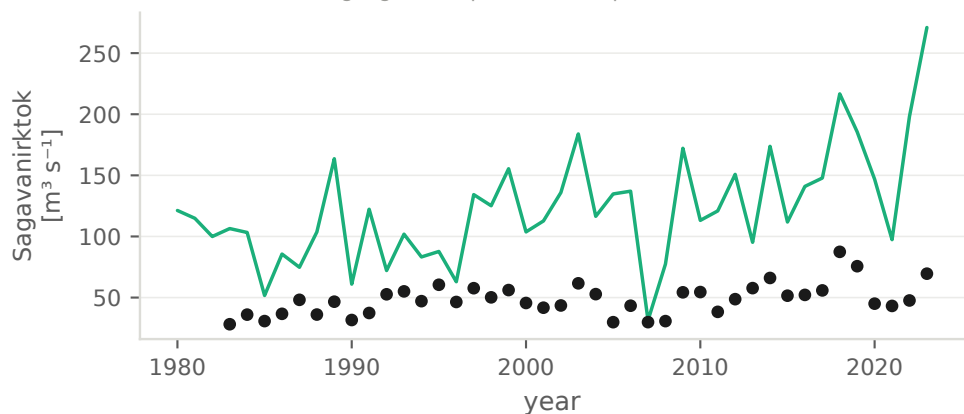
gauged far upstream (Umiat) — understates mouth flow



near tidewater — best-constrained comparison

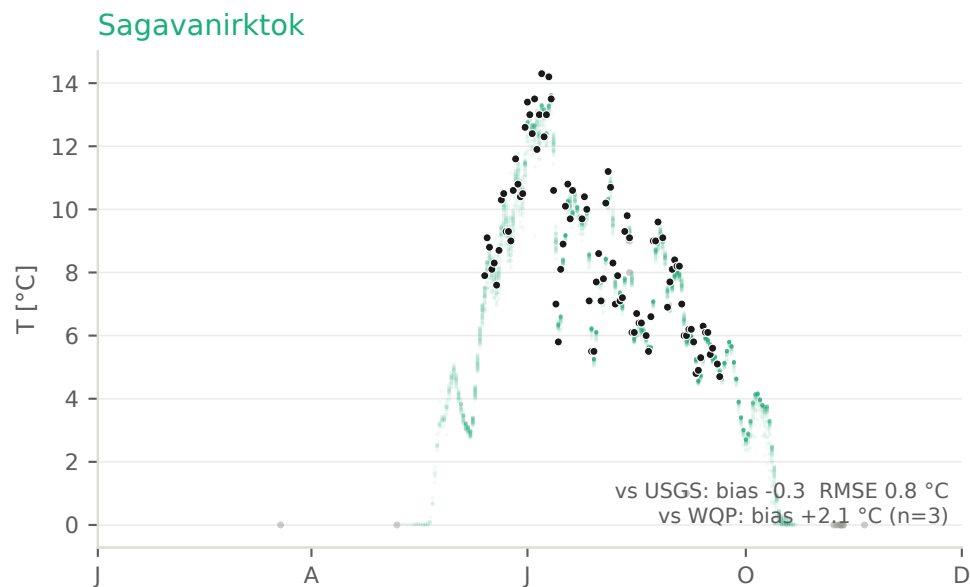
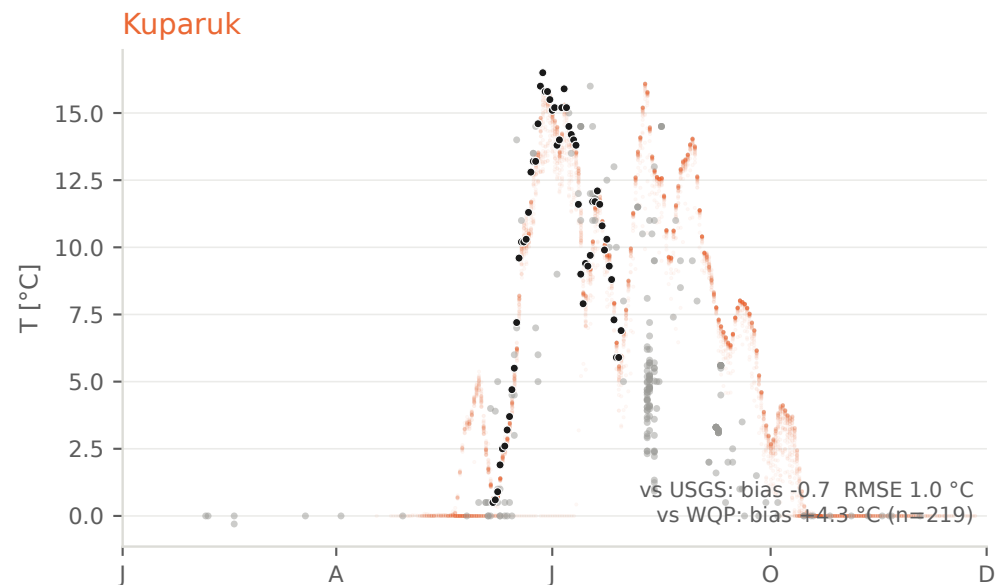
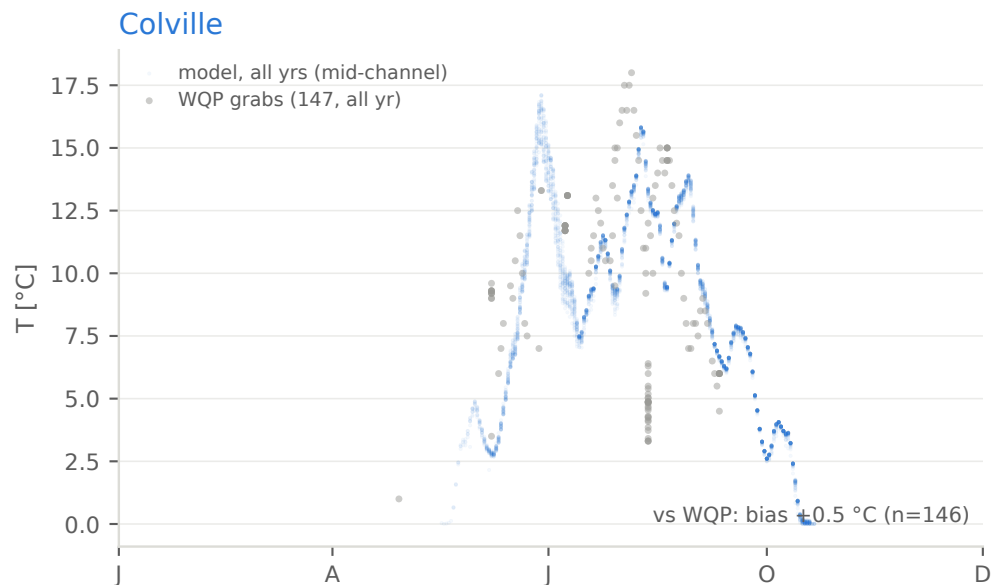


gauged far upstream (Pump Sta 3) — understates mouth flow

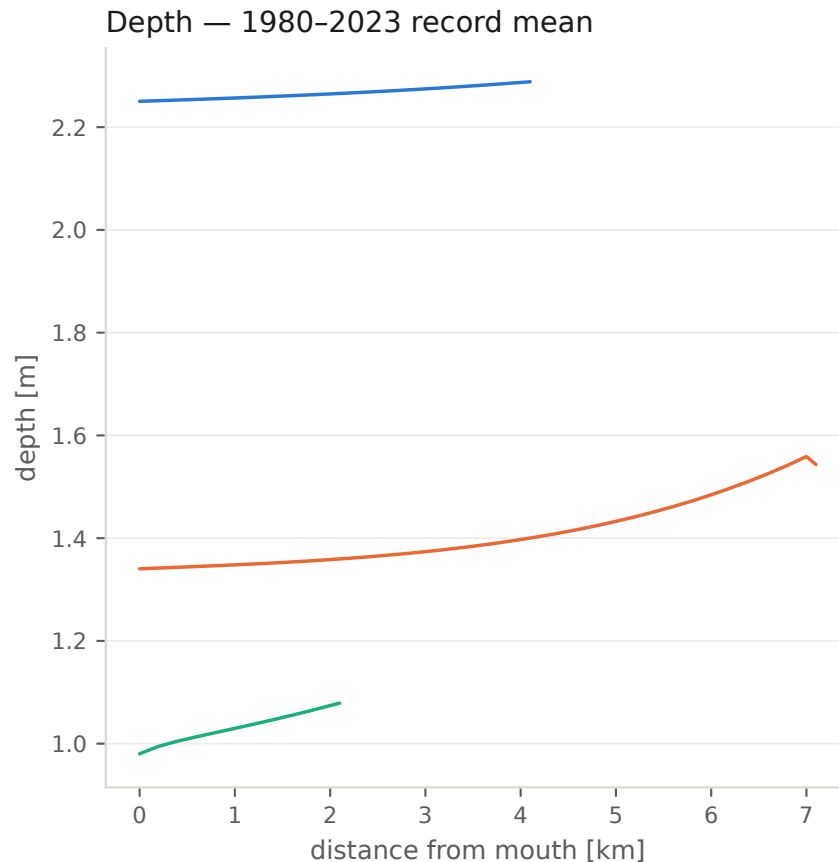
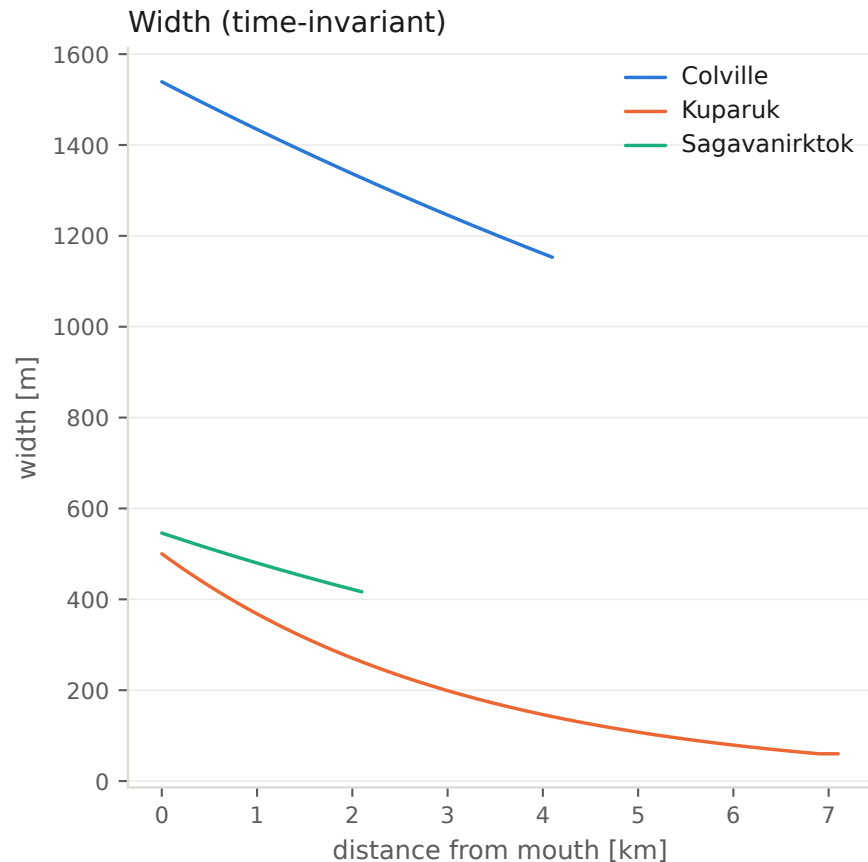


Water temperature — model (all 43 years) vs. observation

Every simulated day, mid-channel, folded to day-of-year and pooled across all 43 post-warmup years (semi-transparent — density shows where the interannual spread is highest)

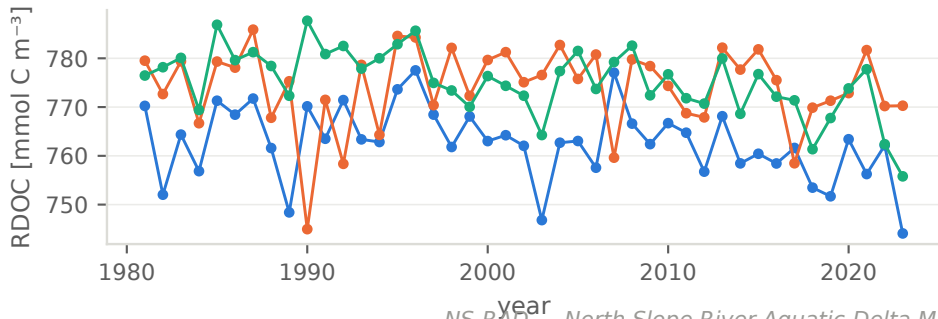
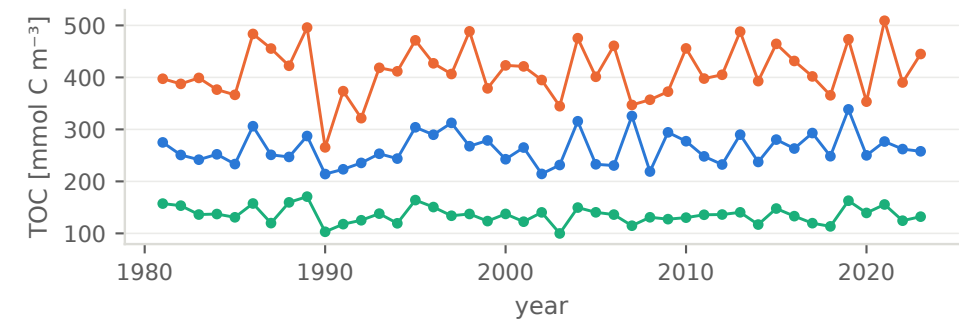
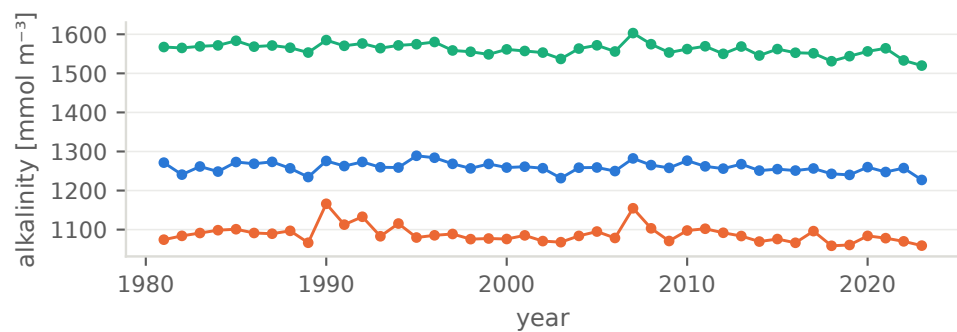
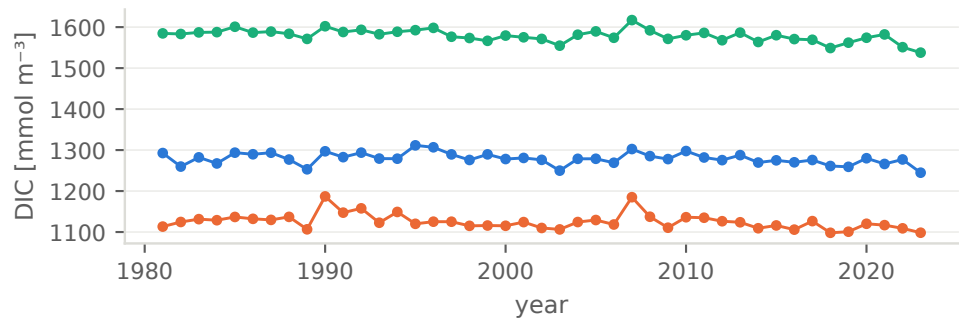
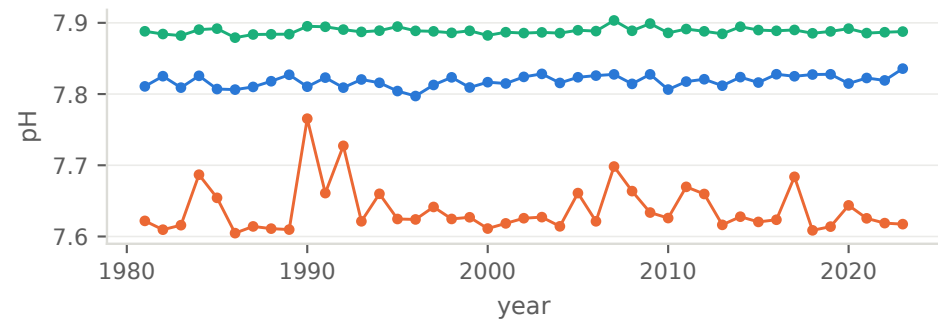
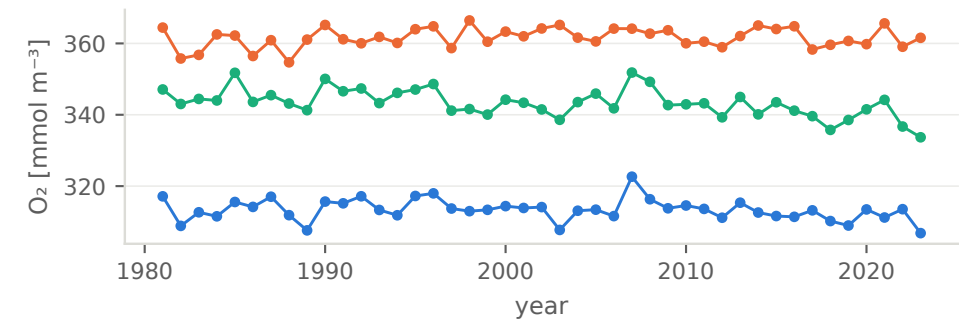
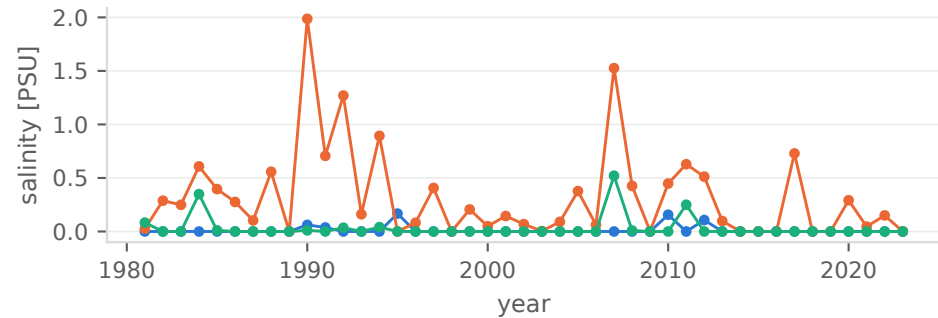
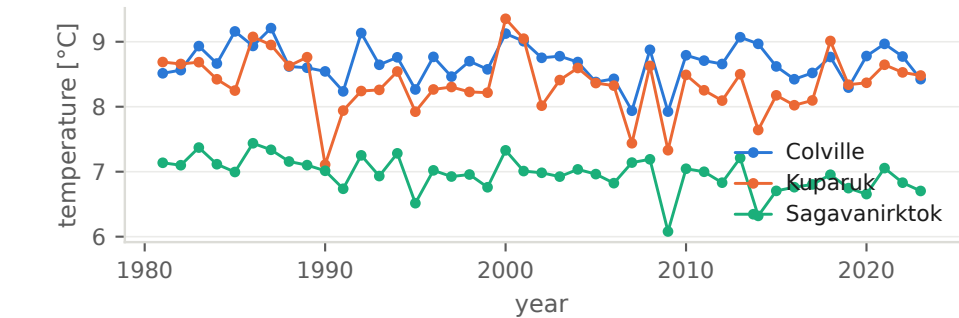


Channel geometry (reference — unchanged from the regular site)



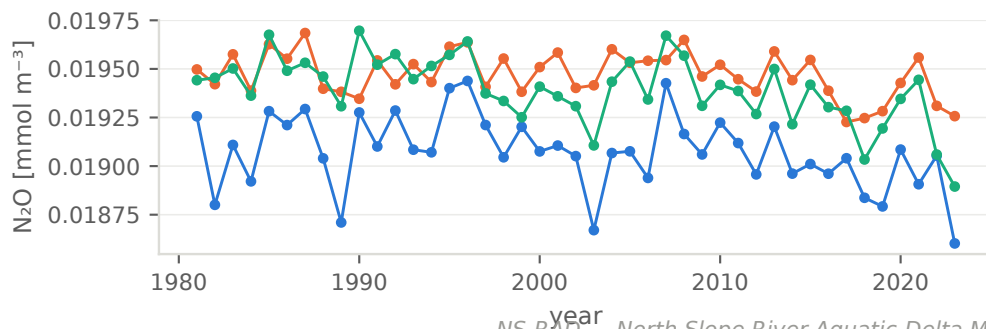
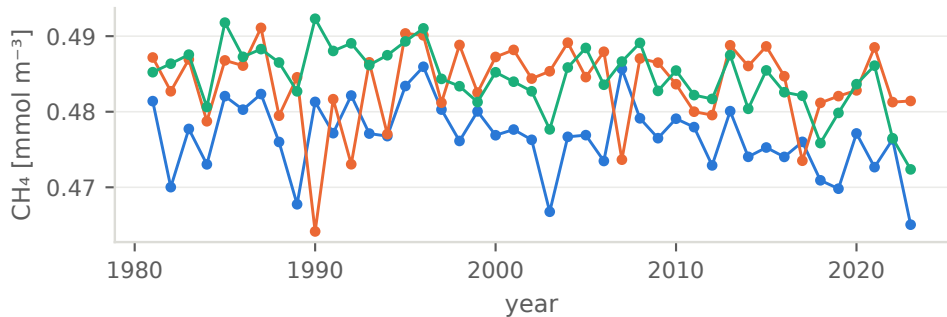
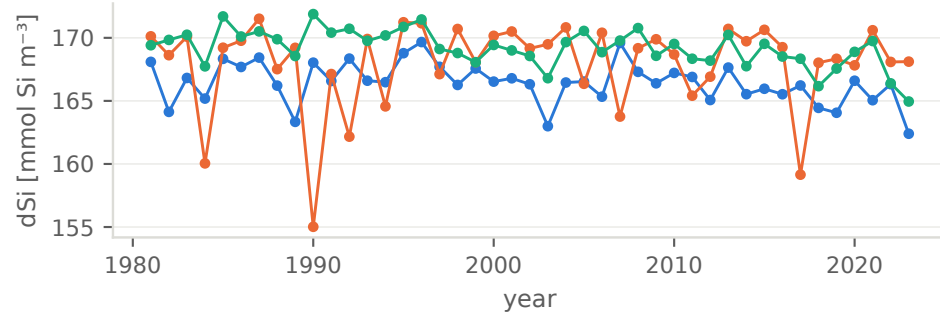
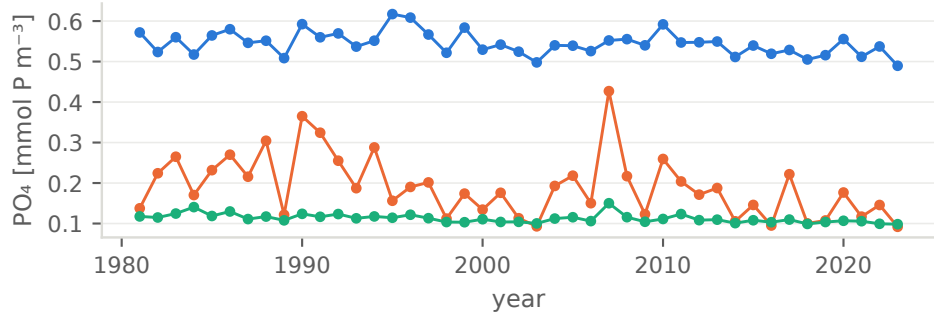
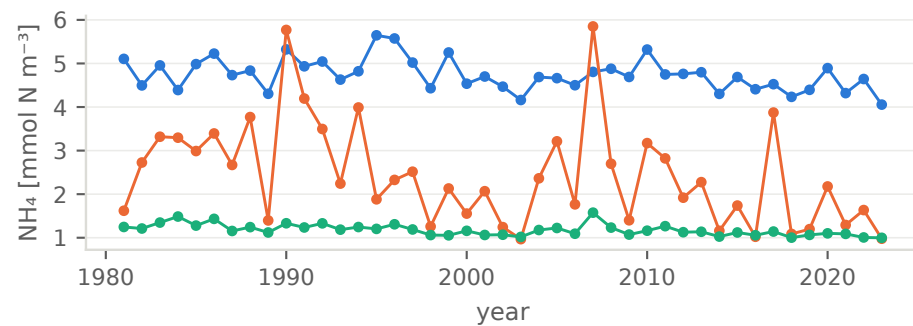
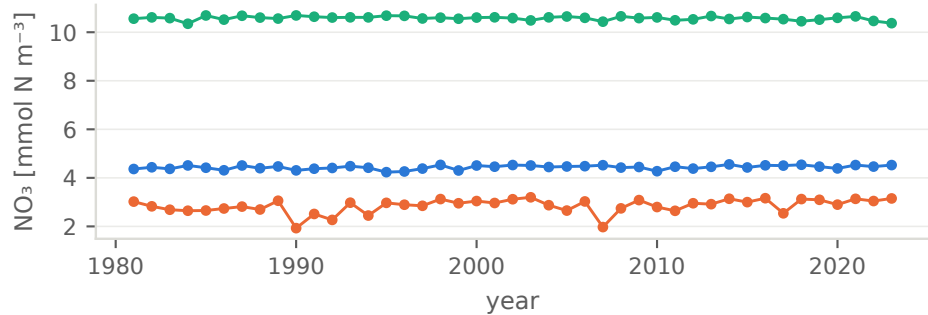
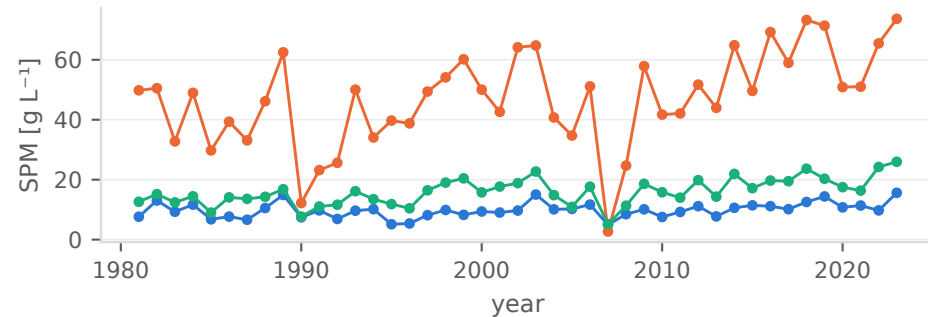
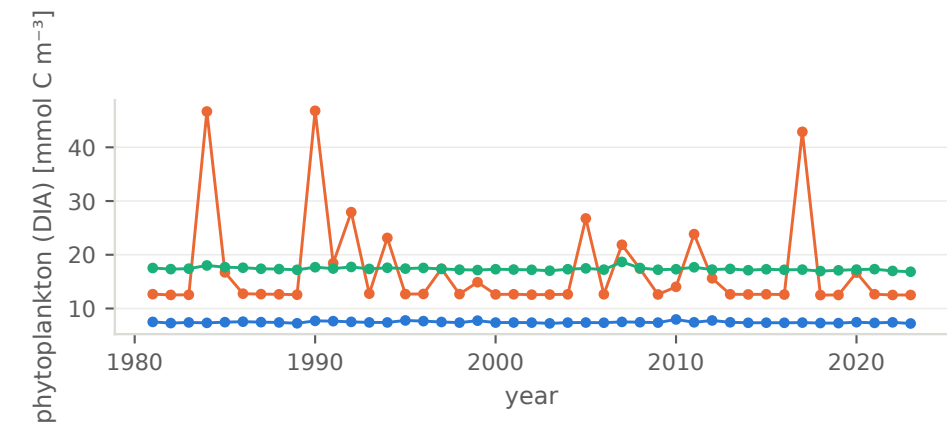
Modelled state variables — annual open-water mean, 1981-2023 (1/2)

Every tracked species (config.py's species registry), spatial mean over open-water cells ($\text{ice_frac} < 0.5$), averaged across each full calendar year — how the



Modelled state variables — annual open-water mean, 1981-2023 (2/2)

Every tracked species (config.py's species registry), spatial mean over open-water cells ($\text{ice_frac} < 0.5$), averaged across each full calendar year — how the



Model response: area-normalized annual budgets, 1981-2023 (1/2)

Every process rate written to output.nc. Same method as ns_rad_diagnostics.pdf's single-year area-normalized page (depth-integrate a volumetric rate, with



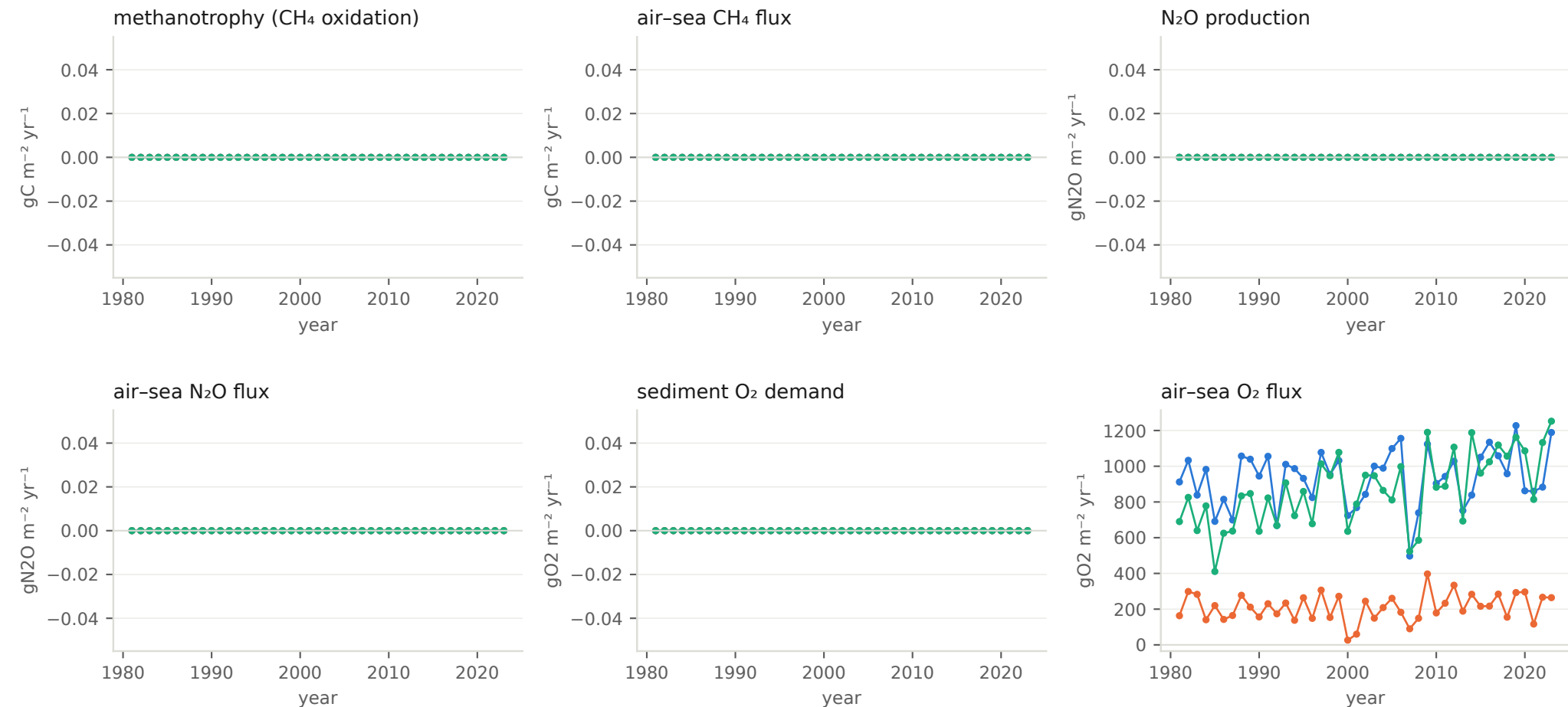
Surface area: Colville 5.481 km^2 · Kuparuk 1.445 km^2 · Sagavanirktok 1.005 km^2

rdoc_ox / photo / ch4_ox / ch4_ex / n2o_prod / n2o_ex / sod are flat zero — config.ARCTIC_BGC is off (default) for these sites, so the Arctic extension reactions never run; RDOC/CH4/N2O still

NS-RAD — North Slope River-Aquatic-Delta Model

Model response: area-normalized annual budgets, 1981-2023 (2/2)

Every process rate written to output.nc. Same method as ns_rad_diagnostics.pdf's single-year area-normalized page (depth-integrate a volumetric rate, with



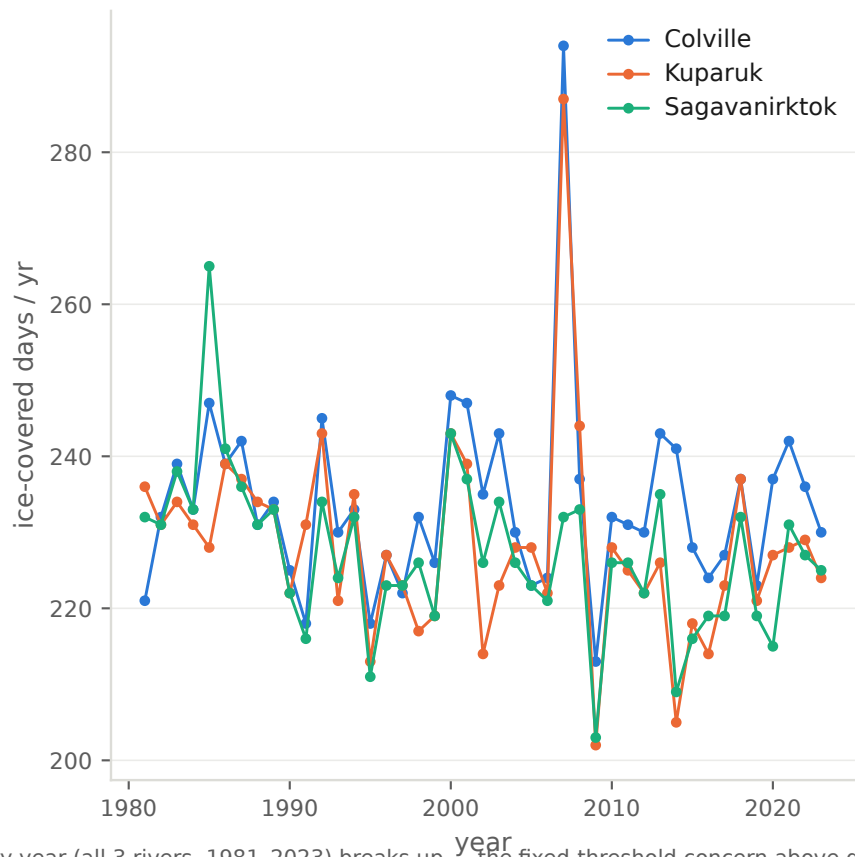
Surface area: Colville 5.481 km² · Kuparuk 1.445 km² · Sagavanirktok 1.005 km²

rdoc_ox / photo / ch4_ox / ch4_ex / n2o_prod / n2o_ex / sod are flat zero — config.ARCTIC_BGC is off (default) for these sites, so the Arctic extension reactions never run; RDOC/CH₄/N₂O still

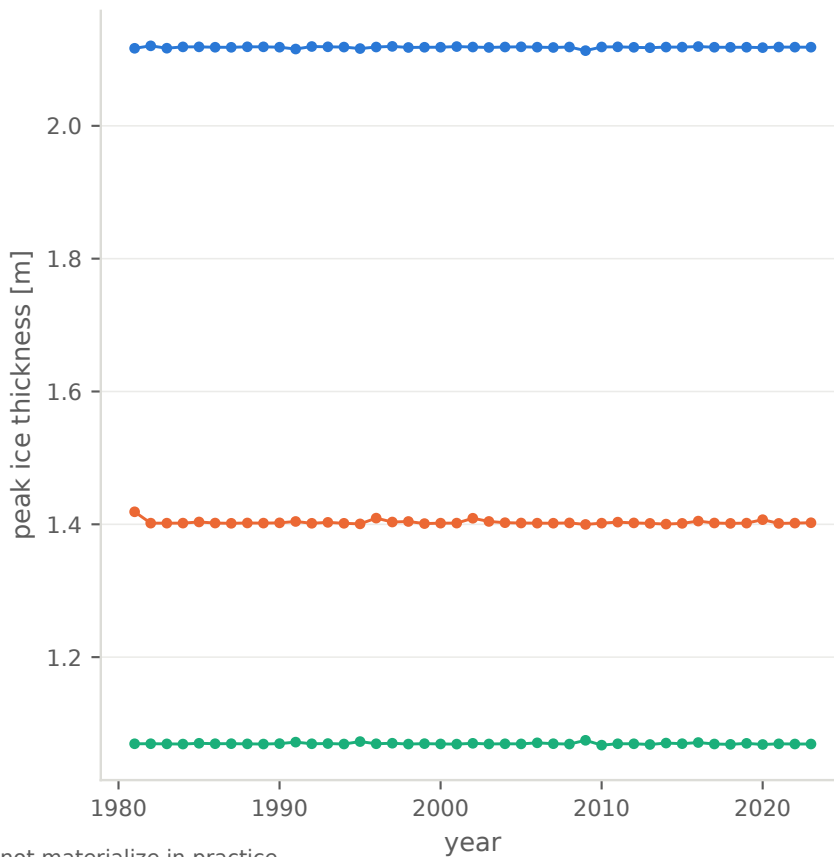
NS-RAD — North Slope River-Aquatic-Delta Model

Ice variability, 1981-2023

The hydraulic breakup mechanism (config BREAKUP_Q_FACTOR=3.0 $\times$ q_ref) makes breakup timing a downstream consequence of the interannual discharge

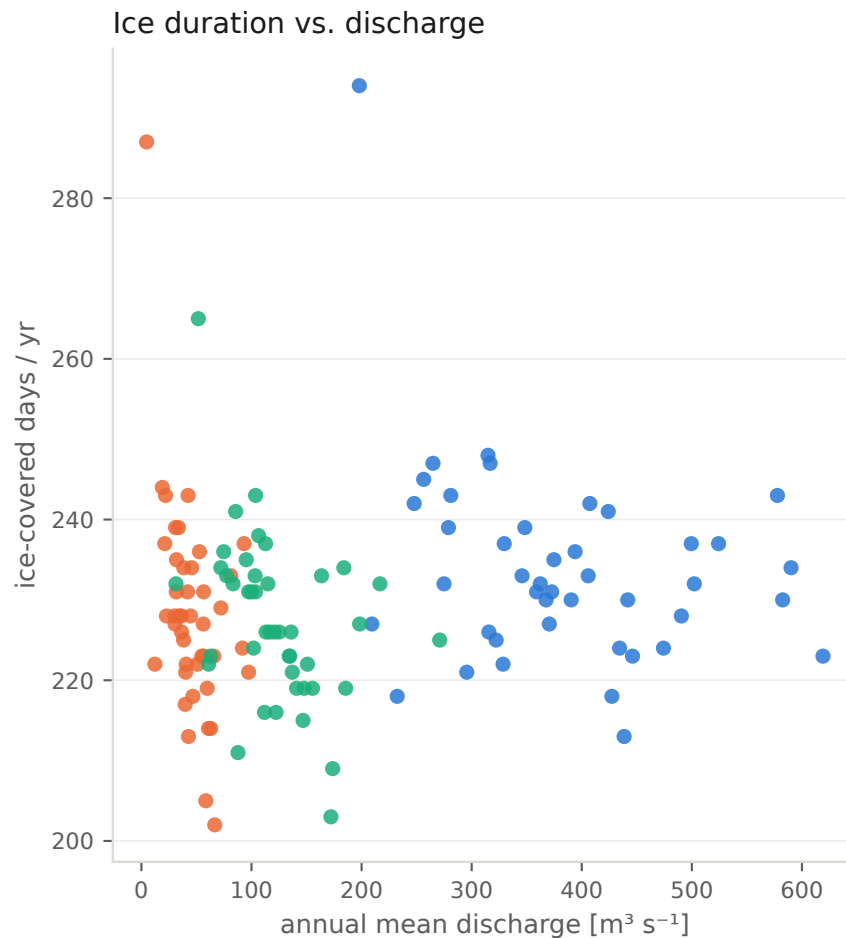
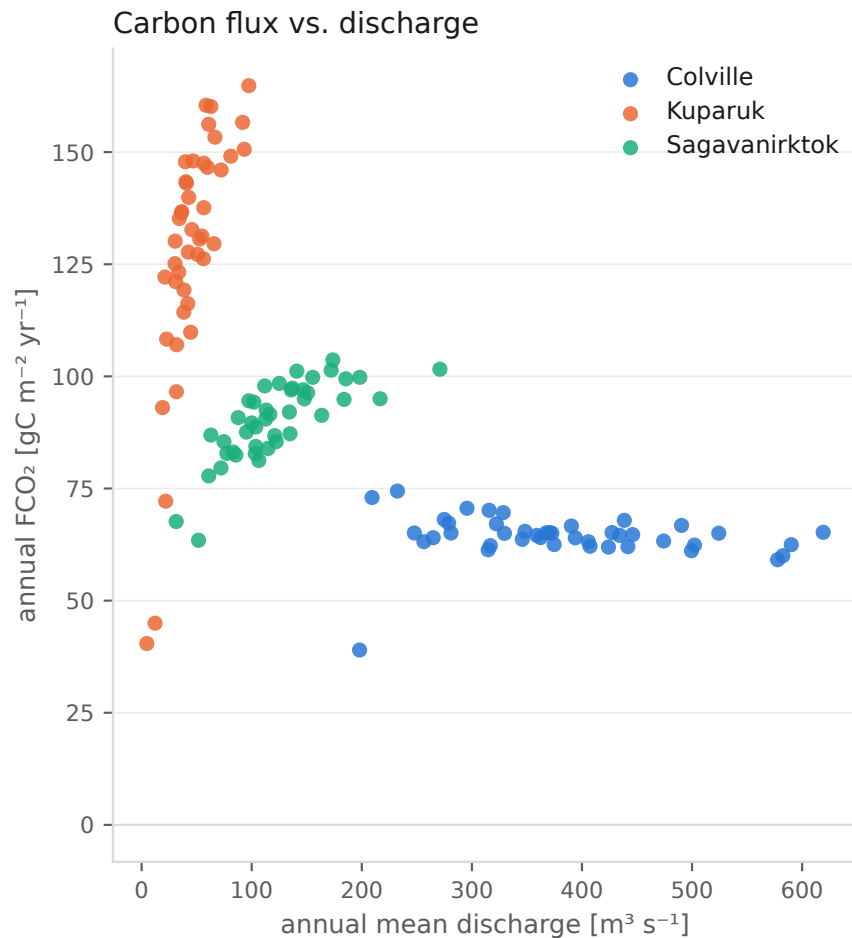


Every year (all 3 rivers, 1981-2023) breaks up — the fixed-threshold concern above does not materialize in practice.



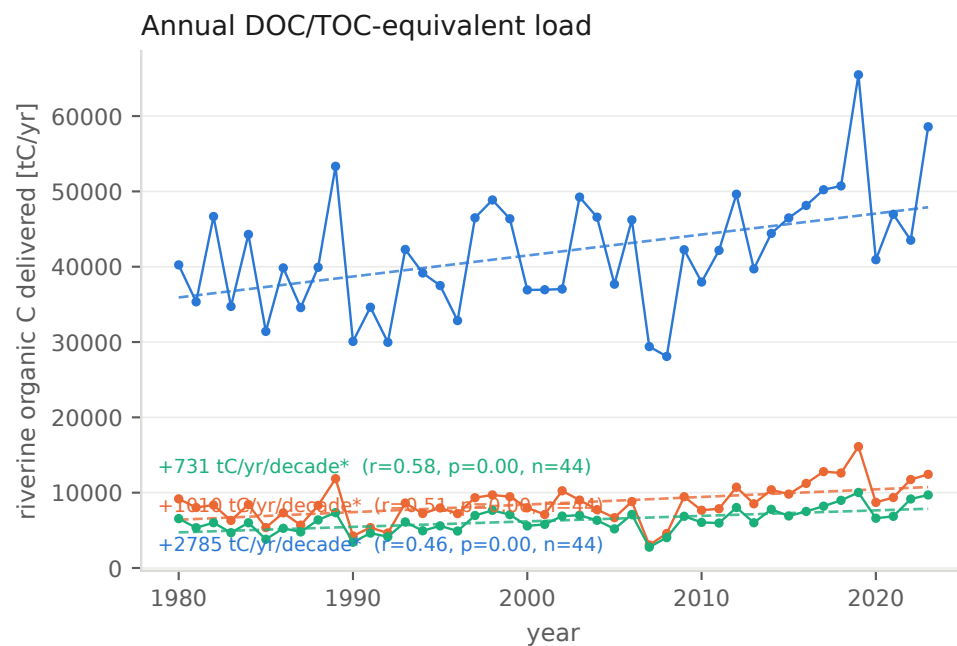
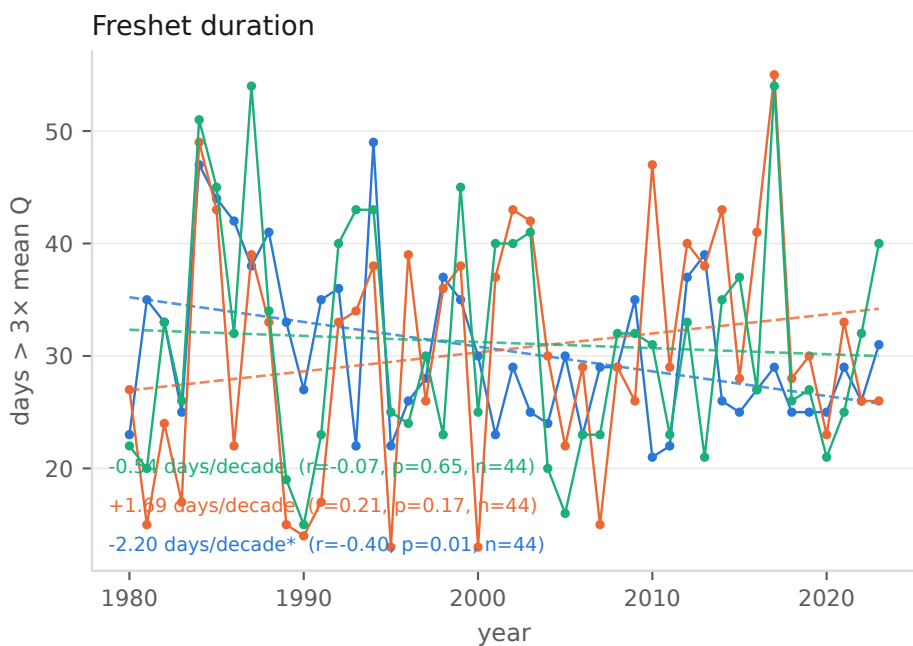
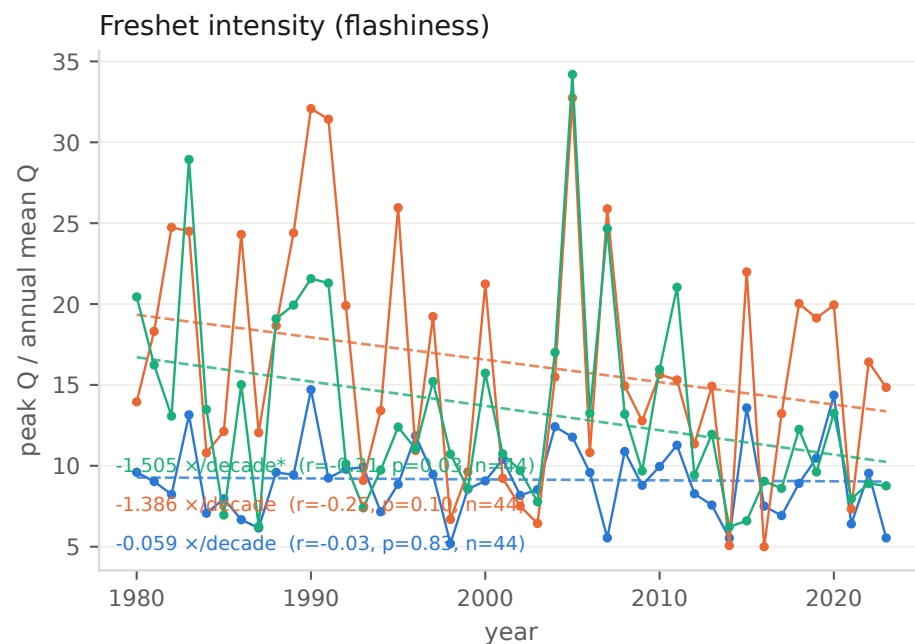
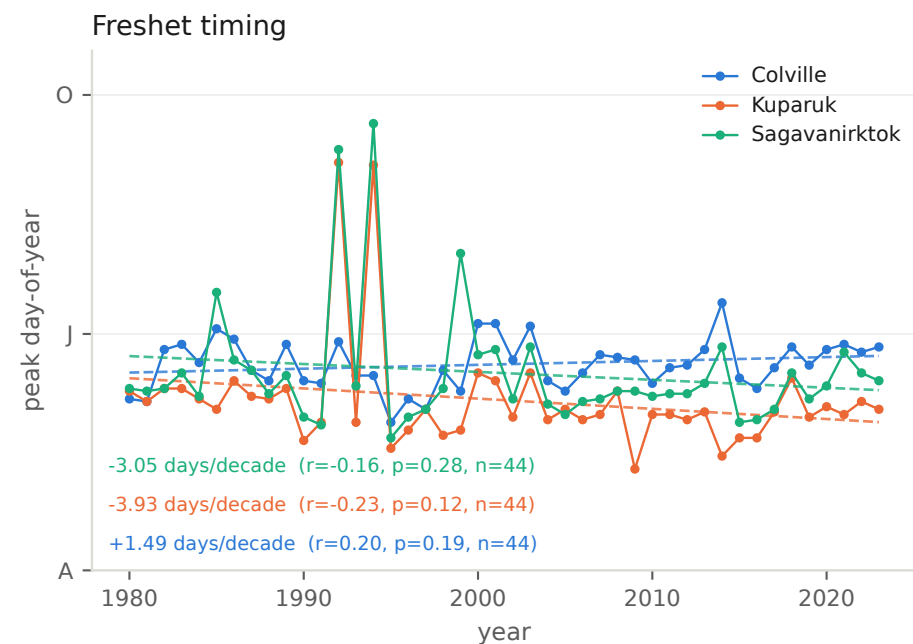
NS-RAD — North Slope River-Aquatic-Delta Model

Does the model's response track the forcing?



Spring freshet characteristics, 1980-2023

Straight from the daily discharge/TOC forcing (all 44 years, no model run needed). 'Freshet' = $Q > \text{BREAKUP_Q_FACTOR}$ (3x) that year's own mean, the SAME threshold t



Does freshet timing/intensity shape downstream ice and carbon flux?

FORM Hypothesis 1: Spring freshet timing and intensity modulate ... sea ice cover and the sign/magnitude of air-sea CO₂ flux. Left: freshet peak day vs. that year's

